LITERATURE SURVEY ON META-LEARNING-BASED HALLUCINATION DETECTION AND RISK PREDICTION IN LARGE LANGUAGE MODELS

DETECTING HALLUCINATIONS IN LLMS USING SEMANTIC ENTROPY (NATURE, 2024)

SUMMARY

- The Problem: Traditional uncertainty measures (like "naive entropy") treat different phrasings of the same correct answer (e.g., "Paris" vs. "It is Paris") as different, leading to misleadingly high uncertainty scores.
- The Solution: The authors propose Semantic Entropy, which clusters generated answers based on their meaning rather than their exact wording.
- How it Works: The method samples multiple answers, uses an LLM or Natural Language Inference (NLI) tool to determine if they entail each other (semantic equivalence), and then calculates entropy over these clusters.
- Key Results: Semantic entropy significantly outperforms baselines across diverse tasks (TriviaQA, BioASQ, math problems) and models (LLaMA 2, Mistral, GPT-4), achieving an average AUROC of 0.790 compared to 0.691 for naive entropy.

RELEVANCE TO THE TOPIC

- This paper provides key foundations for meta-learning–based hallucination detection and risk prediction in large language models:
- Semantic Clustering as a Feature:Semantic cluster distributions offer a meaning-aware alternative to raw token probabilities and can serve as robust input features for meta-learning–based hallucination detection.
- Unsupervised Baseline: As the method requires no labeled hallucination data, it serves as a strong unsupervised benchmark for comparison with meta-learning approaches.
- Risk Evaluation Metric (AURAC): The Area Under the Rejection Accuracy Curve (AURAC) enables explicit evaluation of hallucination risk through selective abstention.
- Black-Box Applicability: The discrete semantic entropy variant supports hallucination detection and risk prediction without access to model internals, enabling use with proprietary LLMs.

ULTRAFEEDBACK: BOOSTING LANGUAGE MODELS WITH SCALED AI FEEDBACK (ICML 2024)

SUMMARY

- Proposes ULTRAFEEDBACK, a large-scale AI-generated feedback dataset to align LLMs without relying on expensive human annotations.
- Introduces a scalable AI feedback pipeline using strong LLMs (e.g., GPT-4) to generate preference judgments across instruction-following, truthfulness, honesty, and helpfulness.
- Constructs over 1 million feedback instances covering diverse tasks and response qualities.
- Trains a reward model (UltraRM) and improves open-source LLM performance using best-of-n sampling and reinforcement learning.
- Demonstrates that AI-generated feedback can effectively replace human feedback for alignment and evaluation at scale.
- Limitations: feedback quality depends on the annotator LLM, and alignment focuses on preference learning rather than explicit hallucination correction.

RELEVANCE TO THE TOPIC

- Provides a scalable supervision signal that can be leveraged for hallucination detection without human-labeled data.
- Complements semantic entropy by enabling learning-based interpretation of uncertainty signals through reward modeling.
- Supports risk-aware evaluation by ranking and rejecting low-quality or unreliable responses.
- Motivates meta-learning approaches that adapt hallucination risk estimation across tasks using AI-derived feedback signals instead of fixed heuristics.

KEY TAKEAWAY

ULTRAFEEDBACK enables scalable, learning-based supervision that complements uncertainty-based hallucination detection and supports meta-learning–driven risk prediction.

THE GEOMETRY OF TRUTH: EMERGENT LINEAR STRUCTURE IN LLM REPRESENTATIONS (COLM 2024)

SUMMARY

- Investigates whether large language models internally represent truth and falsehood in a structured manner.
- Uses high-quality true/false datasets and analyzes hidden layer representations of LLMs (e.g., LLaMA-2).
- Shows that truthfulness is linearly separable in representation space, especially in larger models.
- Demonstrates strong cross-dataset generalization of simple linear probes trained to detect truthfulness.
- Provides causal evidence by intervening on identified directions, causing models to flip between true and false outputs.
- Limitations: requires access to internal model activations and focuses on factual truth rather than open-ended hallucinations.

RELEVANCE TO THE TOPIC

- Establishes that hallucination-related properties (truth vs. falsehood) are explicitly encoded in model internals.
- Supports feature-based hallucination detection using latent representations, complementing output-based uncertainty methods like semantic entropy.
- Motivates meta-learning frameworks that learn task-invariant truth signals across datasets and domains.
- Enables risk prediction by combining internal truth representations with uncertainty signals for calibrated hallucination detection.

KEY TAKEAWAY

Truthfulness emerges as a learnable, linearly structured signal in LLM representations, providing a strong foundation for meta-learning–based hallucination detection and risk prediction.

HALLUCINATION IS INEVITABLE: AN INNATE LIMITATION OF LARGE LANGUAGE MODELS (2025)

SUMMARY

- Provides a formal theoretical analysis of hallucination in large language models.
- Defines hallucination as an inconsistency between a computable LLM and a computable ground-truth function.
- Proves, using learning theory, that hallucination cannot be completely eliminated for any computable LLM used as a general problem solver.
- Shows that even with perfect training data, LLMs will hallucinate due to fundamental generalization limits.
- Identifies classes of hallucination-prone tasks and empirically validates theoretical claims.
- Limitations: analysis is conducted in a formalized world and does not directly provide practical detection or mitigation mechanisms.

RELEVANCE TO THE TOPIC

- Provides theoretical justification for focusing on hallucination detection and risk prediction rather than complete elimination.
- Motivates uncertainty-aware and meta-learning approaches that anticipate hallucinations as inevitable events.
- Supports framing hallucination detection as a risk estimation problem instead of binary correctness.
- Strengthens the need for adaptive, task-aware meta-learning models that manage hallucination risk in real-world deployments.

KEY TAKEAWAY

Hallucinations are theoretically unavoidable in LLMs, making detection and risk prediction—rather than elimination—a necessary research direction.

HALLUCINATION DETECTION IN LARGE LANGUAGE MODELS WITH METAMORPHIC RELATIONS (METAQA, 2025)

SUMMARY

- Proposes MetaQA, a zero-resource hallucination detection method based on metamorphic relations (MRs) inspired by software testing.
- Detects hallucinations by mutating prompts and checking whether expected semantic relations between responses are preserved.
- Operates without external knowledge bases and does not require token probabilities or model internals.
- Applicable to both open-source and closed-source LLMs (e.g., GPT-4, Mistral).
- Demonstrates improved precision, recall, and F1-score compared to SelfCheckGPT across multiple QA datasets.
- Limitations: detection depends on the quality of designed metamorphic relations and provides binary detection rather than calibrated risk estimation.

RELEVANCE TO THE TOPIC

- Provides a black-box hallucination detection strategy, complementing uncertainty- and representation-based methods.
- Supports the need for model-agnostic detection signals, which are crucial for meta-learning across heterogeneous LLMs.
- Highlights the limitation of rule-based detection, motivating learning-based risk prediction instead of fixed logical checks.
- Can serve as a baseline detector whose outputs may be integrated into a meta-learning framework for adaptive hallucination risk estimation.

KEY TAKEAWAY

Metamorphic-relation testing enables black-box hallucination detection, motivating meta-learning approaches that generalize detection and risk prediction across models and tasks.

LLM HALLUCINATION DETECTION VIA HIDDEN LAYER TEMPORAL SIGNALS (HSAD, 2025)

SUMMARY

- Proposes HSAD, a hallucination detection framework based on temporal dynamics of hidden-layer representations during autoregressive generation.
- Models hidden states across layers as a temporal signal, inspired by cognitive neuroscience and human deception detection.
- Applies Fast Fourier Transform (FFT) to convert hidden-layer signals into the frequency domain and extract discriminative spectral features.
- Identifies hallucinations as anomalous frequency patterns arising from internal reasoning instability.
- Achieves significant performance gains on benchmarks such as TruthfulQA, outperforming prior state-of-the-art methods.
- Limitations: requires access to internal model activations and is computationally intensive due to signal extraction and transformation.

RELEVANCE TO THE TOPIC

- Demonstrates that hallucinations manifest as dynamic instabilities in internal representations, not just output errors.
- Complements semantic entropy and truth-probe methods by capturing temporal reasoning behavior.
- Provides rich internal features that can be leveraged by meta-learning models for cross-task hallucination detection.
- Motivates risk prediction based on internal signal patterns rather than static confidence scores alone.

KEY TAKEAWAY

Hallucinations leave detectable temporal signatures in LLM hidden states, offering strong internal signals for meta-learning–based detection and risk prediction.

WHY LANGUAGE MODELS HALLUCINATE (KALAI ET AL., 2025)

SUMMARY

- Analyzes hallucinations as a consequence of statistical learning and evaluation objectives in modern LLM training.
- Argues that LLMs are incentivized to guess when uncertain, since benchmark evaluations reward plausible answers over abstention.
- Formalizes hallucination as an error arising from binary classification ambiguity between valid and invalid outputs.
- Shows that hallucinations persist even with high-quality training data due to misaligned scoring and optimization pressures.
- Highlights that hallucination is not mysterious but a systematic outcome of training and evaluation pipelines.
- Limitations: focuses on explaining causes rather than proposing concrete detection or mitigation methods.

RELEVANCE TO THE TOPIC

- Provides a statistical explanation for why hallucination risk must be predicted rather than eliminated.
- Reinforces the need for risk-aware systems that can identify when models are likely guessing.
- Supports uncertainty- and signal-based approaches as inputs to meta-learning–based hallucination risk prediction.
- Motivates evaluation frameworks that reward calibrated abstention, aligning with risk prediction objectives.

KEY TAKEAWAY

Hallucinations arise from training and evaluation incentives that reward guessing, making risk-aware detection and prediction essential.

HALLUCINATION IS INEVITABLE FOR LLMS UNDER THE OPEN WORLD ASSUMPTION (XU, 2025)

SUMMARY

- Reframes hallucination as a consequence of the generalization problem rather than a simple engineering defect.
- Distinguishes between Closed World settings (where training and test distributions align) and Open World settings (unbounded, evolving environments).
- Proves that under the Open World Assumption, hallucinations are theoretically unavoidable, even with improved data or models.
- Classifies hallucinations into correctable (false memorization) and inevitable (false generalization).
- Argues that hallucination should be treated as a structural property of intelligent systems, not merely a bug.
- Limitations: focuses on conceptual and theoretical analysis without proposing concrete detection mechanisms.

RELEVANCE TO THE TOPIC

- Provides a real-world theoretical justification for hallucination risk prediction under open-ended deployment conditions.
- Strengthens the motivation for meta-learning, which enables adaptation across evolving tasks and domains.
- Supports treating hallucination detection as probabilistic risk estimation rather than absolute correctness checking.
- Reinforces the need for adaptive, task-aware hallucination management in practical LLM systems.

KEY TAKEAWAY

Under open-world conditions, hallucination is unavoidable, making adaptive detection and risk prediction a fundamental requirement.

SURVEY OF HALLUCINATION IN NATURAL LANGUAGE GENERATION (ACM CSUR, 2023)

SUMMARY

- Presents a comprehensive survey of hallucination in Natural Language Generation (NLG) across tasks such as summarization, dialogue, QA, data-to-text, and machine translation.
- Defines and categorizes hallucinations into intrinsic (contradicting source input) and extrinsic (contradicting external facts).
- Reviews existing evaluation metrics, benchmarks, and mitigation strategies, including factuality checks, retrieval augmentation, and training objectives.
- Highlights that most existing methods are task-specific, resource-dependent, or rely on external knowledge bases.
- Identifies gaps in generalizable detection, risk-aware evaluation, and cross-task robustness.
- Limitations: focuses on surveying existing methods and does not propose new detection or prediction models.

RELEVANCE TO THE TOPIC

- Provides a structured taxonomy that contextualizes uncertainty-, representation-, and learning-based hallucination detection methods.
- Highlights the lack of adaptive, task-agnostic hallucination risk prediction, motivating meta-learning approaches.
- Supports the need for unified signals that generalize across tasks rather than task-specific heuristics.
- Serves as a baseline reference to position meta-learning–based hallucination detection and risk prediction as a novel direction.

KEY TAKEAWAY

Existing hallucination research is fragmented across tasks, motivating a unified, meta-learning–based approach for detection and risk prediction.

SIMPLE FACTUALITY PROBES FOR DETECTING HALLUCINATIONS IN LONG-FORM GENERATION (EMNLP 2025)

SUMMARY

- Proposes a lightweight factuality probe that detects hallucinations using hidden states from a single LLM generation, avoiding expensive multi-sample methods.
- Shows that LLM hidden representations encode fine-grained factuality signals even in long-form outputs spanning thousands of tokens.
- Trains simple linear and tree-based probes on hidden states to predict claim-level hallucinations.
- Achieves performance comparable to semantic entropy and self-consistency methods while using up to 100× fewer FLOPs.
- Demonstrates strong cross-model generalization, where probes trained on smaller open-source models transfer to large proprietary models.
- Limitations: requires access to internal hidden states and supervised probe training data.

RELEVANCE TO THE TOPIC

- Provides an efficient internal hallucination signal suitable for large-scale and long-form generation scenarios.
- Complements uncertainty-based methods by enabling single-pass hallucination detection, crucial for real-time risk estimation.
- Supplies structured internal features that can be combined with other signals (semantic entropy, black-box checks) in a meta-learning framework.
- Supports calibrated hallucination risk prediction by enabling fine-grained, claim-level confidence estimation.

KEY TAKEAWAY

Under open-world conditions, hallucination is unavoidable, making adaptive detection and risk prediction a fundamental requirement.

DETECTING HALLUCINATIONS IN SCIENTIFIC CLAIMS USING PROMPTING AND INTERNAL STATES (SCIHAL, ACL 2025)

SUMMARY

- Proposes a hallucination detection framework for scientific claim verification as part of the SciHal 2025 Challenge.
- Combines prompting strategies (zero-shot, few-shot, and chain-of-thought) with internal hidden-state representations of LLMs.
- Extracts hidden states from LLM outputs and trains an auxiliary classifier to categorize hallucination types at the claim level.
- Operates on a curated dataset containing real user queries, retrieved scientific abstracts, LLM-generated responses, and human-annotated claims.
- Achieves competitive performance on claim-level hallucination classification across multiple scientific domains.
- Limitations: domain-specific to scientific content and relies on supervised labels and prompt engineering.

RELEVANCE TO THE TOPIC

- Demonstrates the effectiveness of hybrid hallucination signals, combining prompting behavior with internal model representations.
- Highlights how domain context influences hallucination patterns, motivating task-adaptive learning.
- Provides supervised hallucination signals that can serve as training data or auxiliary features for meta-learning frameworks.
- Reinforces the need for generalizable risk prediction, as domain-specific detectors do not easily transfer across tasks.

KEY TAKEAWAY

Hybrid prompting and hidden-state features improve scientific hallucination detection, motivating meta-learning approaches for task-adaptive hallucination risk prediction.

HALUEVAL: A LARGE-SCALE HALLUCINATION EVALUATION BENCHMARK FOR LLMS (2023)

SUMMARY

- Introduces HaluEval, a large-scale benchmark for evaluating hallucinations in large language models across multiple tasks.
- Contains over 35,000 human-annotated samples, including general user queries and task-specific data for QA, dialogue, and summarization.
- Defines hallucinations as content that conflicts with source information or cannot be verified by factual knowledge.
- Uses a sampling-then-filtering strategy to automatically generate challenging hallucinated responses, followed by human verification.
- Provides span-level annotations, enabling fine-grained evaluation of hallucination detection methods.
- Limitations: benchmark focuses on recognition rather than prediction and relies on induced hallucinations that may differ from real-world deployment settings.

RELEVANCE TO THE TOPIC

- Serves as a standard evaluation benchmark for comparing hallucination detection and risk estimation methods.
- Enables quantitative assessment of task-agnostic hallucination detection models, including meta-learning approaches.
- Highlights the difficulty of hallucination recognition even for strong LLMs, motivating risk-aware prediction frameworks.
- Provides labeled data that can support meta-learning across tasks by exposing diverse hallucination patterns.

KEY TAKEAWAY

HaluEval provides a large-scale, fine-grained benchmark that exposes the limitations of current hallucination detectors and motivates adaptive, learning-based risk prediction.

FELM: BENCHMARKING FACTUALITY EVALUATION OF LARGE LANGUAGE MODELS (NEURIPS 2023)

SUMMARY

- Introduces FELM, a comprehensive benchmark for evaluating factuality in LLM-generated text across five domains: world knowledge, science & technology, mathematics, reasoning, and writing.
- Provides fine-grained, span-level annotations with explicit error types, explanations, and supporting or contradicting references.
- Moves beyond Wikipedia-centric evaluation by including diverse real-world domains and reasoning tasks.
- Evaluates both vanilla LLMs and retrieval-augmented models, showing that factuality detection remains challenging even for state-of-the-art systems.
- Emphasizes the need for interpretable and fine-grained factuality evaluators.
- Limitations: focuses on evaluation rather than proposing new hallucination detection or mitigation methods.

RELEVANCE TO THE TOPIC

- Provides a multi-domain evaluation framework suitable for assessing hallucination detection and risk prediction models.
- Enables analysis of generalization across domains, which is critical for meta-learning approaches.
- Highlights the gap between detection performance and real-world reliability, motivating calibrated hallucination risk estimation.
- Supports benchmarking meta-learning models that combine multiple hallucination signals.

KEY TAKEAWAY

FELM exposes the limitations of current factuality evaluators across domains, reinforcing the need for adaptive, meta-learning–based hallucination risk prediction.

AUTOHALL: AUTOMATED HALLUCINATION DATASET GENERATION FOR LLMS (2025)

SUMMARY

- Proposes AutoHall, an automated framework for constructing model-specific hallucination datasets without manual annotation.
- Leverages existing fact-checking datasets to generate hallucinated references and determine factual inconsistencies automatically.
- Reveals that different LLMs exhibit distinct hallucination rates and types, highlighting model-dependent hallucination behavior.
- Introduces a zero-resource, black-box hallucination detection method based on self-contradiction among generated references.
- Provides empirical analysis showing hallucination prevalence of approximately 20–30% across modern LLMs.
- Limitations: focuses primarily on factual hallucinations and dataset construction rather than general-purpose risk prediction.

RELEVANCE TO THE TOPIC

- Addresses the data scarcity problem for training hallucination detection and risk prediction models.
- Enables scalable generation of task- and model-specific training data, which is crucial for meta-learning frameworks.
- Highlights variability in hallucination patterns across models, motivating adaptive, model-aware risk prediction.
- Provides datasets and detection signals that can support meta-learning across heterogeneous LLMs.

KEY TAKEAWAY

AutoHall enables scalable, model-specific hallucination datasets, supporting meta-learning approaches that adapt hallucination risk prediction across models and tasks.

FACTOOL: TOOL-AUGMENTED FACTUALITY DETECTION FOR GENERATIVE AI (2023)

SUMMARY

- Proposes FACTOOL, a task- and domain-agnostic framework for detecting factual errors in LLM-generated text using tool augmentation.
- Decomposes long-form generations into claims, generates verification queries, retrieves evidence via external tools (e.g., web search, scholar, code interpreters), and performs reasoning-based verification.
- Supports multiple tasks, including knowledge-based QA, code generation, mathematical reasoning, and scientific writing.
- Demonstrates improved factuality detection compared to task-specific baselines, especially in complex, long-form outputs.
- Emphasizes evidence-backed verification and interpretable decision-making.
- Limitations: depends on external tools and retrieval quality, increasing latency and reducing applicability in constrained or offline settings.

RELEVANCE TO THE TOPIC

- Illustrates a mitigation-oriented approach that complements detection by grounding outputs in external evidence.
- Highlights trade-offs between accuracy and efficiency, motivating lightweight risk prediction when tool use is costly.
- Provides structured signals (claim validity, evidence consistency) that can be integrated as features in meta-learning frameworks.
- Reinforces the need for adaptive risk estimation to decide when tool-based verification is necessary.

KEY TAKEAWAY

Tool-augmented verification improves factuality but incurs cost and latency, motivating meta-learning–based risk prediction to selectively trigger mitigation.

CHAIN-OF-VERIFICATION (COVE): REDUCING HALLUCINATIONS VIA SELF-VERIFICATION (2023)

SUMMARY

- Introduces Chain-of-Verification (CoVe), a generation-time method where the model verifies its own outputs to reduce hallucinations.
- The model first produces a draft answer, then plans verification questions, answers them independently, and finally revises the response based on verification results.
- Demonstrates consistent reductions in hallucinations across QA, list-based queries, and long-form generation.
- Shows that factored verification (preventing conditioning on the initial draft) improves correction effectiveness.
- Highlights the role of deliberate self-checking in improving factual accuracy during generation.
- Limitations: increases inference cost and latency; effectiveness depends on the model's verification ability.

RELEVANCE TO THE TOPIC

- Represents a mitigation-focused approach complementary to post-hoc detection.
- Highlights the cost–accuracy trade-off, motivating risk prediction to decide when verification is necessary.
- Produces intermediate verification signals that can serve as features for meta-learning–based risk estimation.
- Reinforces the need for adaptive control policies that trigger mitigation selectively based on predicted hallucination risk.

KEY TAKEAWAY

Self-verification can reduce hallucinations but is costly, motivating meta-learning–based risk prediction to apply verification selectively.

METACHECKGPT: MULTI-TASK HALLUCINATION DETECTION USING META-MODELS (SEMEVAL 2024)

SUMMARY

- Proposes MetaCheckGPT, a meta-learning–based hallucination detection framework developed for SemEval-2024 Task 6 (SHROOM).
- Combines outputs and uncertainty signals from multiple large language models into a meta-regressor that predicts hallucination likelihood.
- Supports both model-aware (using logits / probabilities) and model-agnostic (black-box) detection settings.
- Evaluated across multiple generation tasks including definition modeling, machine translation, and paraphrase generation.
- Demonstrates improved hallucination detection performance compared to single-model and heuristic-based approaches.
- Limitations: relies on supervised task-specific data and focuses on detection rather than calibrated risk estimation.

RELEVANCE TO THE TOPIC

- Directly aligns with meta-learning–based hallucination detection, making it a key reference for this work.
- Demonstrates how meta-models can aggregate heterogeneous hallucination signals across tasks and models.
- Provides empirical evidence that learning to combine signals outperforms fixed detection rules.
- Motivates extending meta-learning from binary detection to continuous hallucination risk prediction.

KEY TAKEAWAY

MetaCheckGPT validates meta-learning as an effective paradigm for hallucination detection, motivating risk-aware extensions for adaptive hallucination prediction.

TRUTHFULQA: MEASURING HOW MODELS MIMIC HUMAN FALSEHOODS (NEURIPS 2022)

SUMMARY

- Introduces TruthfulQA, a benchmark designed to evaluate whether language models generate truthful answers rather than plausible but incorrect ones.
- Focuses on questions where humans commonly hold false beliefs or misconceptions, making hallucination detection challenging.
- Covers multiple domains including health, law, finance, and general knowledge.
- Evaluates models using both generation-based and multiple-choice settings.
- Shows that larger and more capable models often produce more confident yet incorrect answers.
- Limitations: limited domain coverage and focuses primarily on factual truthfulness rather than broader hallucination types.

RELEVANCE TO THE TOPIC

- Provides early empirical evidence that hallucinations persist despite model scaling.
- Highlights the need for hallucination risk awareness, not just accuracy optimization.
- Motivates uncertainty- and learning-based approaches for hallucination detection.
- Supports framing hallucination as a probabilistic risk phenomenon, aligning with meta-learning–based risk prediction.

KEY TAKEAWAY

TruthfulQA demonstrates that confident falsehoods are common in LLMs, motivating risk-aware hallucination detection beyond accuracy metrics.

SELFCHECKGPT: ZERO-RESOURCE BLACK-BOX HALLUCINATION DETECTION (EMNLP 2023)

SUMMARY

- Proposes SelfCheckGPT, a hallucination detection method based on self-consistency across multiple model generations.
- Detects hallucinations by sampling multiple responses to the same prompt and measuring semantic inconsistency among them.
- Operates in a black-box setting, requiring no access to model internals, training data, or external knowledge bases.
- Applicable across tasks such as question answering, summarization, and long-form generation.
- Serves as a strong baseline for uncertainty-based hallucination detection.
- Limitations: computationally expensive due to repeated sampling and does not provide calibrated risk estimates.

RELEVANCE TO THE TOPIC

- Establishes a baseline uncertainty signal for hallucination detection in black-box LLMs.
- Motivates improved uncertainty estimation methods such as semantic entropy.
- Highlights the need for learning-based aggregation of uncertainty signals rather than fixed inconsistency thresholds.
- Serves as an input signal for meta-learning–based hallucination risk prediction.

KEY TAKEAWAY

Self-consistency reveals hallucination risk in black-box settings, motivating meta-learning approaches for adaptive risk prediction.

PROPOSED RESEARCH DIRECTION

- Hallucinations in Large Language Models are inevitable in open-world settings, making complete elimination impractical.
- Existing detection methods rely on individual signals such as uncertainty measures, internal representations, or black-box consistency checks.
- This work proposes a meta-learning–based framework that learns to adaptively combine multiple hallucination signals across tasks.
- The objective is hallucination risk prediction, enabling selective abstention or verification rather than binary detection.

CORE IDEA

- My work proposes a meta-learning framework to combine multiple hallucination signals for adaptive hallucination risk prediction.
- Learn how to detect hallucinations, not just what signal to use.

THANK YOU